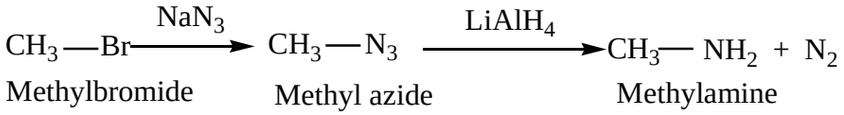
to the formation of quarternary ammonium salt. However, if the process is carried out with excess ammonia, primary amine is obtained as the major product.

The order of reactivity of alkylhalides with amines.

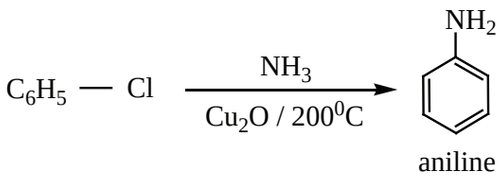
$$RI > RBr > RCl$$

c) Alkyl halide can also be converted to primary amine by treating it with sodium azide (NaN_3) followed by the reduction using lithium aluminium hydride.



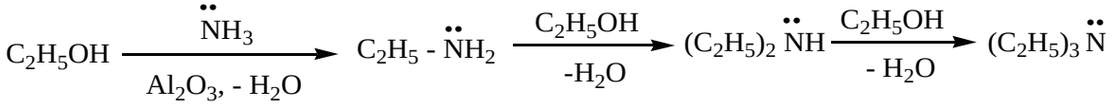
d) Preparation of aniline from chlorobenzene

When chlorobenzene is heated with alcoholic ammonia, aniline is obtained.

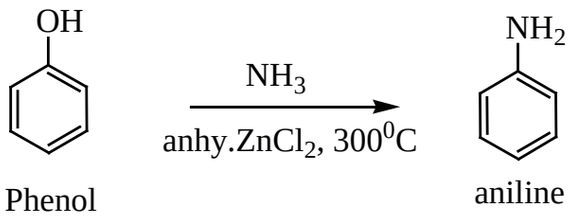


5) Ammonolysis of hydroxyl compounds

a) when vapour of an alcohol and ammonia are passed over alumina, W_2O_5 (or) silica at 400^0C, all types of amines are formed. This method is called **Sabatier – Mailhe** method.



b) Phenol reacts with ammonia at 300^0C in the presence of anhydrous $ZnCl_2$ to give aniline



13.2.4 Properties of amines

1. Physical state and smell

The lower aliphatic amines (C_1-C_2) are colourless gases and have ammonia like smell and those with four or more carbons are volatile liquids with fish like smell.

Aniline and other arylamines are usually colourless but when exposed to air they become coloured due to oxidation.

2. Boiling point

Due to the polar nature of primary and secondary amines, can form intermolecular hydrogen bonds using their lone pair of electorn on nitrogen atom. There is no such H-bonding in tertiary amines.

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